THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the bulb in the tester does not glow with iron nail, it could be because: 1

(a) the nail is coated with paint or rust **(b)** iron is an insulator
(c) nail is a plastic **(d)** cell is stronger

2. A circuit has a cell, a switch, a bulb and a fuse in one loop. If the fuse melts, the bulb: 1

(a) glows brighter **(b)** stays the same **(c)** glows dimmer **(d)** goes off

3. Which statement about a fuse wire is correct? 1

(a) it is made of rubber
(b) it never melts
(c) its melting point is lower than that of the connecting wires
(d) its melting point is higher

4. Which of these should not be done to identify a live wire? 1

(a) switch off main supply **(b)** use a tester
(c) touch it **(d)** use insulated tools

5. A bulb glows when three cells of 1.5 V are connected in series, but not when one cell is reversed. The reason is: 1

(a) the bulb is faulty
(b) the reversed cell reduces the total voltage
(c) the reversed cell adds voltage
(d) wires are copper

6. A circuit tester lights up when connected across a lemon, but not when connected across a dry wooden stick. This suggests: 1

(a) the bulb is wrong **(b)** the lemon is a metal
(c) wood conducts **(d)** the lemon juice conducts

7. Why are high-voltage power lines placed on tall poles and not buried in ordinary ways? 1

(a) to cool **(b)** to avoid contact and shock
(c) to look nice **(d)** to save wire

8. Which of these would stop a tester from lighting? 1

(a) a closed switch **(b)** a clean bulb
(c) a broken connecting wire **(d)** a new cell

9. Why does the filament of a bulb not burn in the bulb? 1

(a) bulb is open **(b)** filament is cold
(c) filament is wet **(d)** the bulb has inert gas or vacuum

10. Why is it unsafe to use an iron with a damaged cord? 1

(a) the cord is heavy **(b)** cord is wet
(c) the iron will cool **(d)** exposed wire can shock

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a bulb connected to a battery of three cells glow brighter than with one cell? 2

12. Why are the bodies of plugs and switches made of plastic? 2

13. Why can a 5 A fuse not be replaced with a 15 A fuse in a lighting circuit? 2

14. Why is a switch connected in series with an appliance? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student wants to make a battery of 9 V using 1.5 V cells. How many cells are needed and how are they joined? 3

16. A fuse of 10 A blew while using many appliances. Explain what to do. 3

17. Explain why batteries in a toy are removed when it is not used for a long time. 3

OR

Why does a bulb connected to a battery of cells in the wrong order not glow? 3

18. Explain how LED bulbs help reduce power plants' pollution. 3

OR

Why do we switch off lights and fans when leaving a room? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an experiment to show how the heating of a wire depends on its length or thickness. 5

OR

Evaluate whether we should use metal or plastic for the body of a hair dryer and the heating plate. 5

20. Explain electrical safety in a school laboratory including wiring, fuse, switches and first aid. 5

OR

Describe how the invention of electric cell and bulb changed life, and the role of safety rules. 5

END OF QUESTION PAPER — Total: 40 Marks